

RAHUL SHARMA, Ph.D.

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Research Interests

X-ray Astronomy, X-ray Binaries, High-energy Astrophysics, Neutron Stars, Cataclysmic Variables, Black Holes, Magnetars, X-ray Pulsars, Ultra-luminous X-ray sources (ULXs), X-ray Polarization, Orbital evolution, Ultra-compact Binaries, GW sources, X-ray Transients, Time-series analysis, XNAV

Research Experience

- Dec 2024 – **  **Post-Doctoral Fellow** – Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune, India.
- Dec 2021 – Nov 2024  **Post-Doctoral Fellow** – Raman Research Institute, Bangalore, India.
- Jan 2021 – Apr 2021  **Research Associate** – IISER, Mohali, India.

Education

- 2014 – 2021  **Ph.D. Physics**, University of Delhi.
Thesis title: *Study of transient X-ray binaries in the time and energy domains.*
Supervisor: Prof. Anjan Dutta.
- 2011 – 2013  **M.Sc. Physics**, University of Delhi.
- 2008 – 2011  **B.Sc. (H) Physics**, Ramjas College, University of Delhi.

Development

- POLIX  Active member of the POLIX science collaboration. This group works on the science goals of the POLIX instrument (built at RRI) onboard *XPoSat* (X-ray Polarimeter Satellite).
- Instrumentation  Development of HK parameters file, participate in the thermo-vac testing of the POLIX instruments, conduct POLIX pipeline testing, Analysis of initial dataset of Crab pulsar, source and background calibration.

Awards and Achievements

National Eligibility Test

-  Cleared CSIR/UGC National Eligibility Test (**NET-JRF**) in Physical Sciences in Dec-2012.

Grants

- 2023  **AHEAD2020 TNA** – INAF-IAPS, Italy.
- 2019  **Cospar PCB** – University of Cagliari, Italy.
- 2016–2019  **UGC-Senior Research Fellowship** – University of Delhi, India.
- 2014–2016  **UGC-Junior Research Fellowship** – University of Delhi, India.

Awards and Achievements (continued)

Allocated Observing Time

- PI  *XMM-Newton* – AO16 (20 ks)
 *AstroSAT* – To3 (80 ks), To5 (40 ks), A14 (60 ks), To5 (345 + 345 + 345 ks)
 *NICER* – ToO (2+22 ks)
 *NuSTAR* – Cycle 10 (60 ks), Cycle 11 (60 ks), ToO (2 x 20 ks)
 *Swift* – ToO (2 x 2 ks)
 **Optical:** *SALT* – 2026-2 (2.9 ks)
- Co-I  *AstroSAT* – Ao2 (24+20+20 ks), To3 (50 ks), To5 (40 ks)

Membership

- ASI  Life Member of the Astronomical Society of India (ASI) - [L2552](#)
- IAU  Junior Member of the International Astronomical Union (IAU) - [42522](#)

Talks & Posters

- Talks  Seminar at Indian Institute of Astrophysics, Bangalore – 20 August, 2026
 Seminar at Raman Research Institute, Bangalore – 18 August, 2026
 *IAU Early Career Astronomer (ECA) Online Discourse Series #30* – 22 May, 2026
[Invited]
 *A Decade of AstroSat Observations: Science Outcomes and Future Prospects*, ISRO, Bangalore – 30 January - 02 February, 2026
 1st International Conference on *COMPACT OBJECT ASTROPHYSICS AND ASTRONOMY (COAA)*, IIT Jodhpur, India – 10-12 December 2025 (**3rd best Oral presentation**)
 National Conference on *REcent Trends in the study of Compact Objects (RETCO-VI): Theory and Observation*, IIT Indore, India – 10-12 March 2025
 *Annual Post-Doctoral Meet (APM 2025)*, Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune – 11 Feb, 2025
 Seminar at Raman Research Institute, Bangalore – 29 Nov, 2024
 *National Space Science Symposium (NSSS 2024)*, Goa University, India – 26 Feb - 01 Mar, 2024
 42nd Meeting of *Astronomical Society of India (ASI 2024)*, IISc Bangalore – Feb 2024
 Seminar at SPASE, IIT Kanpur, India – November 2023
 Seminar at INAF-OAR, Rome, Italy – September 2023
 Seminar at INAF-IAPS, Rome, Italy – September 2023
 *X-ray Universe 2023 Symposium*, Athens, Greece – 13-16 June 2023
 National Conference on *REcent Trends in the study of Compact Objects (RETCO-V): Theory and Observation*, Kodaikanal Solar Observatory, India – 03-05 April 2023
 Astro Journal Club IISER Mohali – February 01, 2023 [Invited]

Talks & Posters (continued)

- 7-years of *AstroSat* Conference, ISRO, Bangalore – September 28-29, 2022 [**Invited**]
- Istanbul University Observatory Astrophysics talks series* (online) – April 2022 [**Invited**]
- Conference on 5 years of AstroSat*, ISRO, Bangalore – January 19-21, 2021
- Seminar* at Department of Physics, University of Cagliari, Italy – 18 February 2020
- Exploring the Universe: Near Earth Space Science to Extra-Galactic Astronomy (EXPUNIV-2018)* at SNBNCBS, Kolkata – 14-17 November 2018
- REcent Trends in the study of Compact Objects (RETCO-III) 2017* at IIST Thiruvananthapuram – June 2017
- Poster *Stellar Variability: Taking the Pulse of the Universe*, IUCAA, Pune – 24-28 November 2025
- X-ray Universe 2023 Symposium*, Athens, Greece – 13-16 June 2023
- XXXVI Meeting of *Astronomical Society of India (ASI 2018)*
- XXXV Meeting of *Astronomical Society of India (ASI 2017)*

Skills

- Languages  Strong reading, writing, and speaking competencies for English and Hindi.
- Coding  HEASOFT, XSPEC, Bash, Python, Gnuplot, C, $\text{\LaTeX}$, ...
- Instruments  *RXTE, NuSTAR, Suzaku, XMM-Newton, Chandra, NICER, Swift, AstroSAT, IXPE, Einstein Probe, POLIX, ...*
- Misc.  Academic research, teaching, training, consultation, $\text{\LaTeX}$ typesetting and publishing.

Services

Referee

- Journal  Provided peer-review of manuscripts submitted to *The Astrophysical Journal (ApJ)*, *Astronomy and Astrophysics (A&A) Letters*, *Monthly Notices of the Royal Astronomical Society (MNRAS)*, and *Journal of Astrophysics and Astronomy (JoAA)*.
- Telescope  Reviewer of observation proposals for the *upgraded Giant Meterwave Radio Telescope (uGMRT)* and *European Southern Observatory (ESO)* Telescopes.

Mentorship

-  Jayatee — Undergraduate Summer Project, University of Delhi (Co-mentor; May–August 2016)
-  Ashrushi S. — Bachelor’s Project, Visiting Student at RRI (May–June 2023)
-  Kinjal Roy — PhD Student, RRI (project mentoring; 2024–2025)
-  Arjun Ramachandran — Bachelor’s Project (Online; 2024–2025)
-  Manish Kumar — PhD Student, RRI (2022–2026)
-  Manish Bishnoi — Master’s Thesis Project, IUCAA–SPPU (2026)
-  Omkar Dalvi — Master’s Project, Fergusson College, Pune (Feb–Apr 2026)

Services (continued)

- Arindamdeep Das — Integrated Master's student, Tezpur University, Assam (May 2026 – **)
- Chehak Singh — B.Tech (Engineering Physics), Delhi Technological University, Delhi (June 2026 – **)

Teaching

- Resource person (Lectures+hands-on) at the *Himalayan School for Training in Astronomy Research (HiSTAR)*, UPES, Dehradun (08-12 June 2026)
- Teaching Assistant (TA) for the course "*Introduction to Astronomy and Astrophysics*" at IUCAA, Pune (2025)
- X-ray Astronomy and Data Analysis lecture for the course "*Astronomy Techniques*" at IUCAA, Pune (2025)
- "Introduction to Cataclysmic Variables and X-ray Pulsar" lecture for the course "*High-energy Astrophysics-I*" at IUCAA, Pune (2025)
- Advising PhD students about X-ray Astrophysics, data analysis, and techniques.
- Resource person (Lecture+hands-on) at the *ASTROCOSMOCON 2023*, SGT University (26-28 October 2023)

Institutional Service

- Member of *IUCAA Committee Against Sexual Harassment (ICASH)*.
- YSOC and LOC member** of the international conference on "*X-ray Polarization of Compact Objects (XPOCO 2027)*" to be organized by IUCAA, Pune (Jan 2027)
- LOC member** of the international conference "*Stellar Variability: Taking the Pulse of the Universe*" organized by IUCAA, Pune (24-28 Nov 2025)
- Tutor and co-organizer** of *High-Energy Data Analysis Workshop* at Raman Research Institute, Bangalore (Aug-Sep 2022)

Miscellaneous

Press Releases & Media Coverage

Dead stars in a six-minute orbit	 IUCAA • Press Information Bureau, MoE • Indian Express • Hindustan Times • NewsBytes • News18 (Hindi) • NDTV (Marathi) • Loksatta (Marathi)
AstroSat detects millisecond bursts	 DST • Times of India • Economic Times • Wionews • Phys.org
Binary eclipses and an exoplanet	 The Hindu
X-ray binary in the LMC	 DST • Phys.org
Science Nugget	 NICER

Outreach

- Coordinator of Weekly Journal Club "*Very Sirius Meeting (VSM)*" at Raman Research Institute (RRI), Bangalore
- Volunteer – *Open Science Day* at RRI and IUCAA for public outreach

Miscellaneous (continued)

- Evaluator at the *National Children's Science Congress (NCSC 2023)* and provided guidance to students
- Volunteered and assisted in organizing the *Visitor's Programme 2016* held at the University of Delhi in March 2016.

References

Available on Request

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Research Publications ([ADS Link](#))

Lead Author Publications

- 1 **R. Sharma**, C. Maitra, F. Haberl, J. Chakraborty, S. Friedrich, Y.-F. Huang, C. Jin, Z. Li, G. Vasilopoulos, Y. Xu, H. Yang, and W. Yuan, “Rapid Orbital Decay in the Ultra-compact Double-degenerate Binary eRASSU J060839.5–704014”, [The Astrophysical Journal Letters](#), **1007**, L7 (2026).
- 2 **R. Sharma**, A. Sanna, and P. Sharma, “Broadband Timing and Spectral Study of Accreting Millisecond X-ray Pulsar SAX J1808.4–3658 during Its 2022 Outburst”, [The Astrophysical Journal](#), **999**, 133 (2026).
- 3 **R. Sharma**, A. Beri, B. Paul, A. Sanna, C. Maitra, and H. Yang, “Probing accretion dynamics and spin evolution in the X-ray pulsar RX J0520.5–6932 during its 2024 outburst”, [Astronomy & Astrophysics](#), **706**, A322 (2026).
- 4 **R. Sharma**, C. Jain, B. Paul, A. Dutta, and V. Rana, “Intriguing nature of AM Her-type candidate CXOU J204734.8+300105”, [The Astrophysical Journal](#), **987**, 117 (2025).
- 5 **R. Sharma**, “Broadband spectral and emission geometry analysis of XB 1916–053 with Chandra and NuSTAR”, [Journal of High Energy Astrophysics](#), **47**, 100376 (2025).
- 6 **R. Sharma**, C. Jain, B. Paul, and A. Beri, “Sidebands to mHz QPOs in 4U 1626–67 in the second spin-down state”, [MNRAS](#), **538**, 1046–1054 (2025).
- 7 **R. Sharma**, M. Mandal, S. Pal, B. Paul, G. K. Jaisawal, and A. Ratheesh, “Probing the energy and luminosity-dependent spectro-timing properties of RX J0440.9+4431 with AstroSat”, [MNRAS](#), **534**, 1028–1042 (2024).
- 8 **R. Sharma**, C. Jain, B. Paul, and T. R. Seshadri, “AstroSat observation of the magnetar SGR J1830–0645 during its first detected X-ray outburst”, [MNRAS](#), **526**, 4877–4884 (2023).
- 9 **R. Sharma**, C. Jain, and B. Paul, “4U 1626–67 returns to spin-down: timing features toe the line”, [MNRAS Letters](#), **526**, L35–L40 (2023).
- 10 **R. Sharma**, A. Sanna, and A. Beri, “AstroSat observation of the accreting millisecond X-ray pulsar SAX J1808.4–3658 during its 2019 outburst”, [MNRAS](#), **519**, 3811–3818 (2023).
- 11 **R. Sharma**, C. Jain, K. Rikame, and B. Paul, “Broad-band mHz QPOs and spectral study of LMC X-4 with AstroSat”, [MNRAS](#), **519**, 1764–1770 (2023).
- 12 **R. Sharma**, A. Beri, A. Sanna, and A. Dutta, “A broad-band look of the accreting millisecond X-ray pulsar SAX J1748.9–2021 using AstroSat and XMM-Newton”, [MNRAS](#), **492**, 4361–4368 (2020).
- 13 **R. Sharma**, C. Jain, and A. Dutta, “Study of the reflection spectra of SAX J1748.9–2021”, [MNRAS](#), **482**, 1634–1639 (2019).
- 14 **R. Sharma**, A. Jaleel, C. Jain, J. C. Pandey, B. Paul, and A. Dutta, “Spectral properties of MXB 1658–298 in the low/hard and high/soft state”, [MNRAS](#), **481**, 5560–5569 (2018).

- 15 R. Sharma, A. Jaleel, C. Jain, B. Paul, and A. Dutta, “Thermonuclear X-ray burst of MXB 1658-298 with NuSTAR”, [Journal of Astrophysics and Astronomy](#), **39**, 16 (2018).

Co-author Publications

- 1 K. Rikame, B. Paul, **R. Sharma**, and V. Jithesh, “Thermonuclear X-ray bursts across the eclipse transitions in the LMXBs EXO 0748-676 and XTE J1710-281”, [MNRAS](#), **544**, 13–23 (2025).
- 2 K. Roy and **R. Sharma**, “Detection of CRSF in Cep X-4 at a low luminosity level”, [New Astronomy](#), **121**, 102435 (2025).
- 3 H. N. Yang, C. Maitra, G. Vasilopoulos, F. Haberl, P. A. Jenke, A. S. Karaferias, **R. Sharma**, et al., “Broadband study of the Be X-ray binary RX J0520.5-6932 during its outburst in 2024”, [MNRAS](#), **536**, 1357–1373 (2025).
- 4 M. Kumar, **R. Sharma**, B. Paul, and V. Rana, “Spectral analysis of Ultra-Luminous X-ray pulsars with models of X-ray pulsars”, [MNRAS](#), **536**, 340–349 (2025).
- 5 K. Roy, **R. Sharma**, H. Manikantan, and B. Paul, “Detection of a cyclotron line in the Be X-ray pulsar IGR J06074+2205”, [Astronomy & Astrophysics](#), **686**, A145 (2024).
- 6 H. Manikantan, B. Paul, **R. Sharma**, P. Pradhan, and V. Rana, “Energy dependence of Quasi-periodic oscillations in accreting X-ray pulsars”, [MNRAS](#), **531**, 530–549 (2024).
- 7 C. Jain, **R. Sharma**, and B. Paul, “A comprehensive study of orbital evolution of LMC X-4: Existence of a second derivative of the orbital period”, [MNRAS](#), **529**, 4056–4065 (2024).
- 8 K. Rikame, B. Paul, **R. Sharma**, V. Jithesh, and K. T. Paul, “Flares during Eclipses of High Mass X-ray Binary Systems Vela X-1, 4U 1700-37, and LMC X-4”, [MNRAS](#), **529**, 3360–3371 (2024).
- 9 A. Devaraj, **R. Sharma**, S. Nagesh, and B. Paul, “Insights into the phase-dependent cyclotron line feature in XTE J1946+274: An AstroSat and Insight-HXMT view”, [MNRAS](#), **527**, 11015–11025 (2024).
- 10 M. Mandal, **R. Sharma**, S. Pal, G. K. Jaisawal, K. C. Gendreau, M. Ng, A. Sanna, C. Malacaria, F. Tombesi, E. C. Ferrara, C. B. Markwardt, M. T. Wolff, and J. B. Coley, “Probing spectral and timing properties of the X-ray pulsar RX J0440.9 + 4431 in the giant outburst of 2022-2023”, [MNRAS](#), **526**, 771–781 (2023).
- 11 A. Beri, **R. Sharma**, P. Roy, V. Gaur, D. Altamirano, N. Andersson, F. Gittins, and T. Celora, “AstroSat and NuSTAR observations of XTE J1739-285 during the 2019-2020 outburst”, [MNRAS](#), **521**, 5904–5916 (2023).
- 12 C. Jain, **R. Sharma**, and B. Paul, “Eclipse Timings of the LMXB XTE J1710-281 : discovery of a third orbital period glitch”, [MNRAS](#), **517**, 2131–2137 (2022).
- 13 C. Jain, A. Yadav, and **R. Sharma**, “Change in spin-down rate and detection of emission line in HMXB 4U 2206+54 with AstroSat observation”, [Journal of Astrophysics and Astronomy](#), **43**, 101 (2022).
- 14 P. Sharma, **R. Sharma**, C. Jain, and A. Dutta, “Timing and spectral analysis of HMXB OAO 1657-415 with NuSTAR”, [MNRAS](#), **509**, 5747–5755 (2022).
- 15 P. Sharma, **R. Sharma**, C. Jain, G. C. Dewangan, and A. Dutta, “Broad-band spectral study of LMXB black hole candidate 4U 1957+11 with NuSTAR”, [Research in Astronomy and Astrophysics](#), **21**, 214 (2021).

- 16 P. Sharma, **R. Sharma**, C. Jain, and A. Dutta, “Broad-band spectral analysis of LMXB XTE J1710-281 with Suzaku”, [MNRAS](#), **496**, 197–205 (2020).
- 17 C. Jain, B. Paul, **R. Sharma**, A. Jaleel, and A. Dutta, “Indication of a massive circumbinary planet orbiting the low-mass X-ray binary MXB 1658-298”, [MNRAS Letters](#), **468**, L118–L122 (2017).

Thesis

- 1 **R. Sharma**, “Study of transient X-ray binaries in the time and energy domains”, in [PhD Thesis, University of Delhi, India](#) (2021).

White Papers and Science Books

- 1 M. Koss and et al. (including **R. Sharma**), “The Advanced X-ray Imaging Satellite (AXIS) Community Science Book”, [arXiv e-prints](#), , [arXiv:2511.00253](#) (2025).

Articles in preprint (ArXiv)

- 1 K. Roy, A. Beri, **R. Sharma**, and P. Charles, “Accretion geometry and spectral evolution in 1A 1118-61: a comparison of the 2009 and 2026 outbursts”, [arXiv](#), [2605.17455](#) (2026).
- 2 M. Kumar, **R. Sharma**, and B. Paul, “Reprocessing of X-rays emission in Ultra-Luminous X-ray sources”, [arXiv](#), [2509.09133](#) (2026).

Non-refereed Article

- 1 K. Roy, **R. Sharma**, H. Manikantan, and B. Paul, “Detection of mHz QPOs in BeXRB pulsar XTE J0111.2-731 (SXP 31.0) in SMC”, [The Astronomer’s Telegram](#), [17204](#) (2025).
- 2 M. Mandal, G. K. Jaisawal, S. Pal, and **R. Sharma**, “Probing the cyclotron resonant scattering feature in the X-ray pulsar 2S 1553-542 with NuSTAR during the 2024 outburst”, [The Astronomer’s Telegram](#), [16859](#) (2024).
- 3 M. Mandal, G. K. Jaisawal, S. Pal, K. Gendreau, **R. Sharma**, J. B. Coley, and C. Malacaria, “NICER observation of Be/X-ray binary pulsar 2S 1553-542 during its 2024 outburst”, [The Astronomer’s Telegram](#), [16846](#) (2024).
- 4 **R. Sharma**, K. Gendreau, Z. Arzoumanian, E. C. Ferrara, P. S. Ray, and A. Sanna, “NICER confirms outburst of RX J0520.5-6932 and detects coherent pulsations”, [The Astronomer’s Telegram](#), [16569](#) (2024).
- 5 **R. Sharma**, “Timing Analysis of intermittent millisecond X-ray pulsar SAX J1748.9-2021”, [Space Research Today](#), **208**, pp. 60–64 (2020).